

Communication Latency in Distributed Inference

THESIS FOLLOW UP PLAN - OCTOBER 6, 2026

Kelsey Penners

Follow Up Plan – goal & constraints

- **Goal:** Strengthen current thesis research into conference-ready paper

Main RQ: How does communication latency and interconnect quality affect tensor-parallel inference performance, scaling efficiency, and energy consumption?

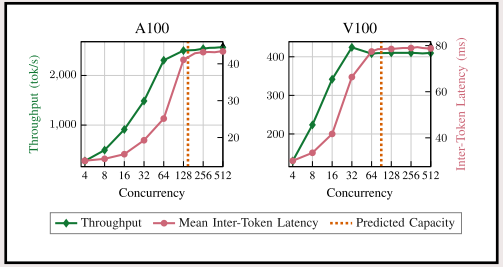
- **Constraints:**
  - *Timeline:* 2 months
  - *Deadlines:* ISC-HPC → Abstract due Dec 14, paper Dec 21
  - *Hardware:*
    - shy-fec (A100-40GB up to TP=4)
    - spike (B200-180GB potentially TP=8)
    - or the cloud
    - deprioritize funnotch (V100-32GB)

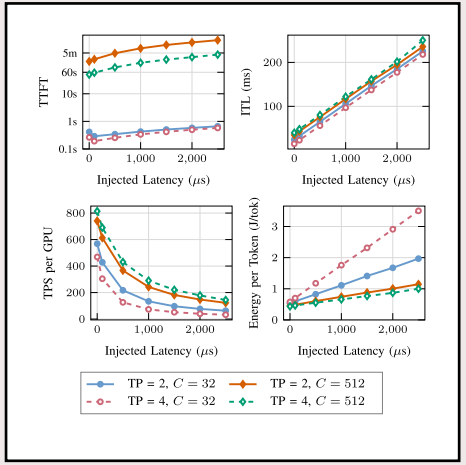
Thesis Recap

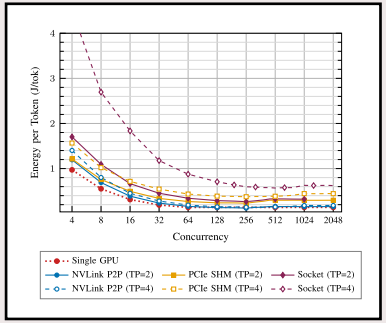
- *Setup*: vLLM on A100 and V100, Llama 1B,8B,13B, TP=1,2,4
- *Experimental controls*: transport protocol (P2P, SHM, Socket) and AllReduce latency injection
- *Key findings*:
  - Optimal TP config depends on hardware topology, not just memory and compute
  - **Transport fallback** and additional **AllReduce latency** incur significant costs to performance
  - Microbenchmark predictions underestimate real transport cost

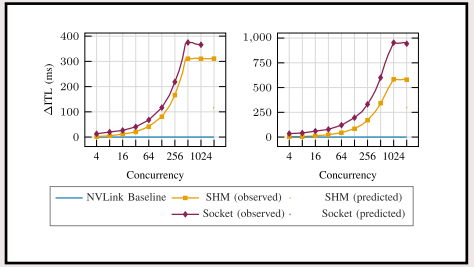
Most Interesting Results

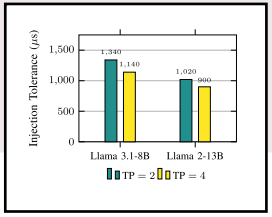
- KV cache saturation behavior
- How performance changes with injected AllReduce latency
- Underestimation when using `nccl-tests` to predict transport fallback degradation
- SLO Tolerance: operations-facing headroom number











Potential Next Steps

- Options for improvement over the next two months (8/9 weeks, ~ 8 hours / week)

Option	Description
`nsys` profiling	Obtain real AllReduce duration and counts per step, might help explain underprediction gap
Repeated runs	Improve rigor by rerunning injection and transport runs
Larger models	Benchmark models with more parameters (e.g., Llama 70B, Qwen 32B, Mistral)
B200 measurements	Measure larger models, up to TP=8, P2P on/off, latency injection generalization

From Spike

- *Resources to request:* up to a Node of 8x B200-180GB GPUs
- *Potential Experiments*: Benchmark larger dense model that need the extra room, TP= 1,2,4,8; P2P on/off, latency injection
- *What I'll have to do* → maybe a week of actual runtime
  - build environment + first test runs,
  - TP=2/4/8 x P2P on/off across a few concurrency points
  - Latency injection for some configurations
- *Hangs on:* whether controlling NCCL transport and latency injection works with server permissions and newer hardware

(Rough) Proposed Timeline

Week	Date	Focus
1	Oct 12	Spike test, experiment scripts, nsys setup
2	Oct 19	nsys profiling on A100, larger model (qwen 32B or llama 70B)
3	Oct 26	
4	Nov 2	Refine AllReduce model with nsys data
5	Nov 9	B200 experiments
6	Nov 16	
7	Nov 23	Full draft done, ready for feedback
8	Nov 30	Revised final draft incorporating comments

Thank you!